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Big Data Analytics in Spark

Group 29's Project

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The Data

1. Italian elections National and regional voting records spanning multiple election cycles. Rich in categorical and numerical structure, ideal for large-scale aggregation and pattern discovery.

2. NYC Yellow Taxi trip records covering fares, routes, pickup/dropoff zones, and timestamps. High volume, high variety, built for machine learning and network analysis.

The 4 V's

1. Volume: millions of election records across decades of Italian voting history, and 350M+ NYC taxi trips

2. Velocity: real-time taxi trip scoring via Spark Structured Streaming, predicting fare and duration as trips come in

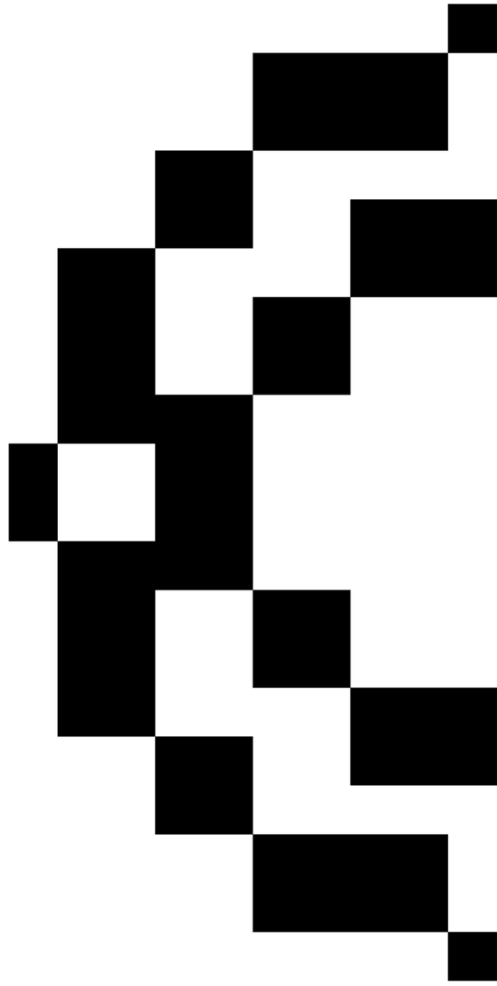
3. Variety: multiple file formats and inconsistent naming conventions across both datasets

4. Veracity: corrupted files, typos, and invalid values across both datasets required extensive cleaning before the data could be trusted

Elections Insights

DataFrames & SparkSQL Let us query structured voting data like a database but at scale, identifying regional patterns, turnout shifts, and party performance across hundreds of constituencies in seconds.

RDDS was tested but DataFrame and SparkSQL proved faster and more efficient structured election data

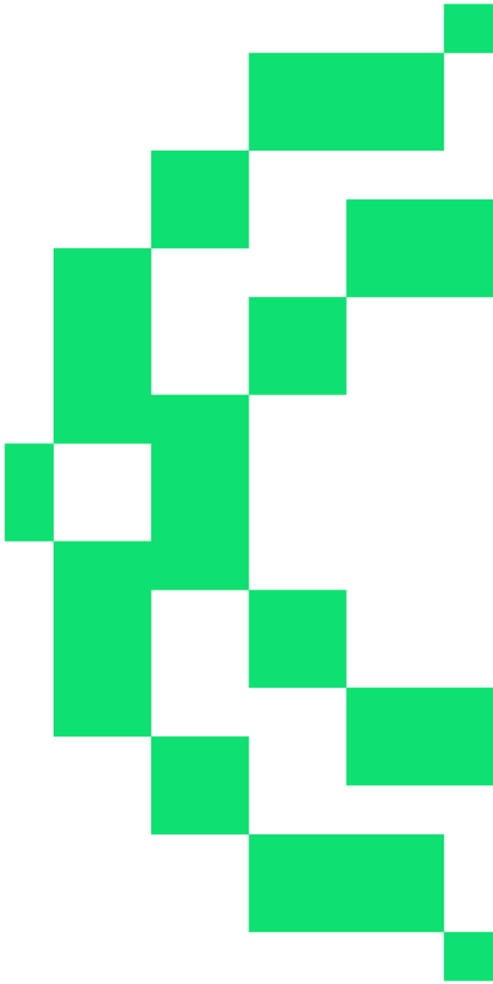


Taxi insights

Pipelines: Spark's Pipeline API processed and transformed the entire dataset in a single distributed pass, ensuring the same logic would scale to millions of real trips without rewriting any code

Machine learning: Spark MLlib trained Random Forest across partitioned data in parallel, something impossible with sklearn on data at this scale

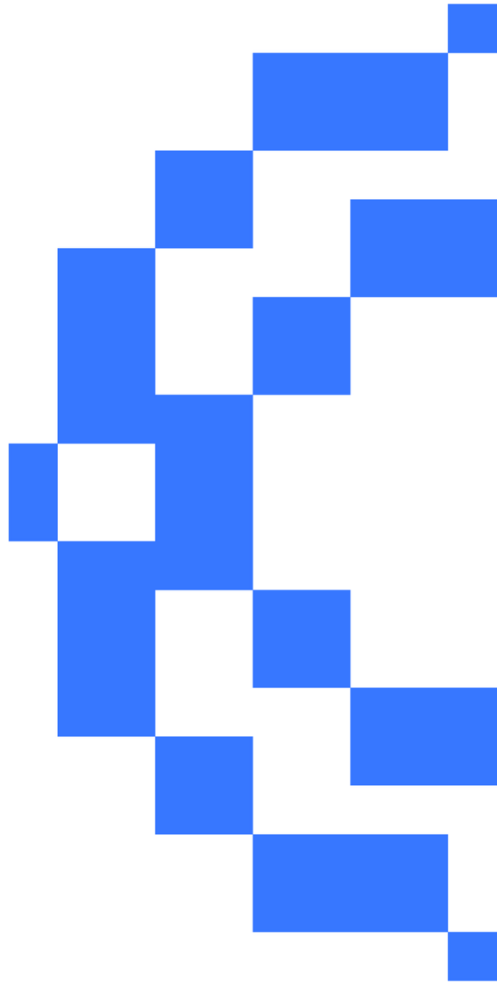
GraphFrames: Modelled pickup and dropoff zones as a network of connected nodes. Spark's distributed graph processing identified the busiest, most connected zones, and underserved areas across the city.



Tools developed

Dashboard: for interactive map and charts letting the client explore any election result by region, party, and turnout at a glance

Streaming: Simulated a real-time trip prediction pipeline, the models can score incoming trips as they arrive, as would happen in a live passenger booking app.



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Conclusion

By adopting Spark, the client gains a single platform that scales with their data.

- What took hours now runs in minutes, across any data volume
- Live predictions delivered as trips happen, not after
- Any team member can explore years of data through the dashboard

References

- NYC Yellow Taxi Trip Data — NYC Taxi & Limousine Commission. kaggle.com/datasets/elemento/nyc-yellow-taxi-trip-data
- Italian Electoral Data — Ministero dell'Interno (1948–2025). elezioni.interno.gov.it
- Apache Spark / PySpark — spark.apache.org/docs/latest/api/python